ModelIt!

Student Activity Pack

Can We Colonize Mars?

Modeling Life Support Systems and Planetary Engineering Challenges

Grade 12L3 | NGSS HS-ESS2-4 | Systems and System Models

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What do you already know about conditions on Mars that make it hostile to human life? List at least three specific challenges.**

2. **If you had to keep 100 people alive in a sealed building for one year with no outside help, what systems would you need?**

3. **Why is it so much harder to sustain life on Mars than on the International Space Station, which has been continuously inhabited since 2000?**

4. **What does 'terraforming' mean to you, and do you think it is scientifically realistic within the next century?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Solar Energy Input
- Atmospheric CO2 Availability
- Subsurface Ice Reserves
- Radiation Exposure Level
- Oxygen Production Rate
- Water Recycling Efficiency
- Food Production Capacity
- Habitat Structural Integrity
- Waste Processing Rate
- Colony Viability Index

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Optimal Conditions

Solar Energy: Maximum | Ice Reserves: Full | Radiation: Shielded | All systems nominal

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Dust Storm Crisis

Solar Energy: -90% for 90 days | Ice Reserves: Full | All systems on battery backup

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Aging Colony

Solar panels at 70% efficiency | Water recycling at 90% | Habitat integrity degraded 15%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Psychological Stress Index — The cumulative mental health burden on colonists from isolation, confinement, communication delays with Earth, and the knowledge that rescue is impossible — psychological breakdown is historically the most common failure mode in extreme isolation environments

Regolith Toxicity Level — The concentration of perchlorates and other toxic compounds in Martian soil that must be removed before the regolith can be used for construction or agriculture — untreated Mars soil is toxic to human skin, lungs, and thyroid function

Communication Delay Impact — The 4-24 minute one-way signal delay between Mars and Earth that prevents real-time troubleshooting — the colony must be capable of diagnosing and repairing any system failure independently without waiting for Earth-based engineering support

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Mars has no breathable atmosphere, no liquid water on its surface, lethal radiation levels, and temperatures that average negative sixty degrees Celsius. Given these extreme conditions, is it scientifically feasible to engineer a self-sustaining human colony on Mars, or are we chasing science fiction? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate why energy is the master variable that controls all other life support systems on Mars?**

2. **What does the dust storm scenario reveal about the minimum redundancy required for colony survival, and how does this change your view of the engineering challenge?**

3. **Why is the Colony Viability Index a better measure of success than any single life support metric, and what threshold would you set for 'sustainable'?**

4. **How would adding psychological stress or regolith toxicity to your model change the predicted colony viability, and which do you think is more dangerous?**

5. **What are the limitations of modeling a Mars colony with ten components when the real system would involve thousands of interacting variables?**

8 STEM Challenge: Design a Mars Colony Life Support Architecture

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YOUR ENGINEERING MISSION

Design an integrated life support system for a 100-person Mars colony that can survive a 6-month global dust storm without resupply from Earth, specifying redundancy strategies for each critical subsystem.

THE SCENARIO

SpaceX and NASA have selected your engineering team to design the life support architecture for the first permanent Mars settlement. The colony must support 100 people indefinitely. Your design must address energy generation and storage, atmospheric management, water recycling, food production, radiation protection, and waste processing. The critical test: your system must survive a six-month global dust storm that reduces solar input by 95% without any resupply from Earth.

Engineering Design Thinking

1. WHAT IS YOUR PRIMARY AND BACKUP ENERGY GENERATION STRATEGY AND HOW MUCH STORAGE CAPACITY DOES YOUR DESIGN INCLUDE?

2. HOW DOES YOUR WATER RECYCLING SYSTEM ACHIEVE THE EFFICIENCY YOUR MODEL SHOWS IS NECESSARY FOR SURVIVAL?

3. WHAT SPECIFIC RADIATION SHIELDING APPROACH DOES YOUR HABITAT USE AND WHAT IS THE PREDICTED CREW EXPOSURE LEVEL?

4. HOW WOULD YOU TEST YOUR LIFE SUPPORT ARCHITECTURE ON EARTH BEFORE COMMITTING 100 PEOPLE TO MARS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Aerospace engineers designing life support systems for NASA and SpaceX earn $85,000-$160,000/year. Planetary scientists studying Mars geology and resources earn $70,000-$130,000/year. Bioregenerative life support engineers who design closed-loop ecological systems earn $80,000-$145,000/year. Astrobiology researchers studying the potential for life beyond Earth earn $65,000-$125,000/year.